



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS: X
SUBJECT: COMPUTER APPLICATIONS

TOTAL MARKS: 40
DATE: 8.7.24

Instructions:

Attempt all questions. The intended marks for questions are given in brackets []. This paper consists of three printed pages.

Question 1

Choose the correct answer from the choices given:

[6×1=6]

i) The process of encapsulating a value within an object and changing a primitive type into a wrapper class is known as _____

- a) converting b) unboxing c) wrapping d) boxing

ii) What is the output of the given expression:

```
int a= -1, b=0, c=3;
```

```
System.out.print(--a - b++ * c-- %2);
```

- a) -3 b) -2 c) 0 d) -1

iii) Assertion(A): If the number of iteration is fixed, it is recommended to use the for loop.

Reason(R): The number of iterations are not fixed, it is advisable to use while or do while loop.

- (a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)
(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)
(c) Assertion (A) is true and Reason (R) is false
(d) Assertion (A) is false and Reason(R) is true

iv) _____ method uses compile time or early binding whereas _____ method uses runtime or dynamic binding.

- a) static, non static b) pure, impure c) non static, static d) impure, pure

v) Wrapper class are present in _____ package.

- a) java.io b) java.lang c) java.util d) java.wrapper

vi) What is the final value of 'ctr' after the iteration process given below executes?

```
int ctr=0;
for(int i=1;i<=5;i++)
ctr++;
for(int j=1;j<=5;j+=2)
++ctr;
```

- a) 6 b) 8 c) 7 d) 9

Question 2

a) Write a statement in java to convert a number stored in a string variable 'n' to double data type. [2]

b) Predict the output: [2]

```
class pg1
{ static public int fun(int x, int y)
{ y-=x; return y; }
public static void main()
{ int a=3, b=7;
b=fun(a,b);
a=fun(b,a);
System.out.println(a + " " +b); } }
```

c) Predict the output: [2]

```
class abc
{ void main()
{ Integer n=new Integer(235);
String s= "235";
int p=Integer.parseInt(s);
p=p+5;
System.out.println(f);
System.out.println(p); }}
```

d) Show the difference between call by value and call by reference with the help of an example? [2]

e) What will be the final value of 'p', when x = -2, y = 4: [2]

```
p = Math.pow(x++ + ++x - --y + x--, --y)
```

f) Convert the given code snippet to equivalent if else construct:

[2]

```
switch(obj)

{ case 'a': System.out.print("Platform Independent");

  case 'b': System.out.print("object oriented");

  default: System.out.print("wrong choice"); }
```

g) The following program segment calculates the norm of a number, norm of a number is the square root of sum of squares of all digits of a number.

For example: The norm of 68 is 10. $(6 \times 6 + 8 \times 8) = 100$, square root of 100 is 10.

Fill in the blanks with appropriate java statements:

[2]

```
void norm(int n)

{ int d, s=___ ;

  while( ____ )

  { d=n%10;

    s = ____;

    n=n/10; }

  System.out.print("norm =" + ____); }
```

Question 3

[10]

Design a class to overload the method display(.....) as follows:

1. void display() — prints the 10th character after the alphabetical letter 'n'.
2. void display(char s, char c) — generate all the alternate letters from 's' to 'c'.
3. int display(char st) — checks and print whether 'st' is a special characters or not. Return 1 if 'st' is a special character otherwise return 0.

Question 4

[10]

Design a menu-driven program to display the following as per the user's choice. Accepting the choice as 'A' to check for Abundant Number and 'B' to check for ugly number.

(a) An Abundant number is a number for which the sum of its proper factors is greater than the number itself.

Example: Consider the number 12. Factors of 12 = 1, 2, 3, 4, 6

Sum of factors = $1 + 2 + 3 + 4 + 6 = 16$.

As $16 > 12$ so 12 is an Abundant number.

(b) An ugly number is a positive integer whose prime factors are limited to 2, 3, and 5.

Example 1: Input n = 6

Output = true

Explanation: $6 = 2 * 3$

Example 2: Input n = 14

Output = false

Explanation: 14 includes 7 as prime factor